CLINICAL · EAR

Ménière's disease

Ménière's disease is a disorder of the inner ear fluids: attacks of spinning vertigo lasting hours, with fluctuating low-tone hearing loss, roaring tinnitus and a feeling of fullness. It is a diagnosis made by criteria after other causes are excluded, and most patients are controlled without ever needing an operation.

AT A GLANCE

- Episodic vertigo, fluctuating low-frequency loss, tinnitus and fullness.
- Diagnosis by criteria, after excluding others.
- Salt reduction and medicines control most.
- Intratympanic gentamicin next; labyrinthectomy if hearing is useless.
- Bárány 2015: definite needs attacks of 20 min–12 h with documented low-frequency loss; betahistine no better than placebo (BEMED).

1 CHAPTER ONE Overview

Episodes of vertigo, an ear that fills and fades.

1.1 Definition

An idiopathic disorder of the inner ear characterised by *endolymphatic hydrops*: distension of the endolymphatic system. It produces the classic tetrad of episodic vertigo, fluctuating sensorineural hearing loss, tinnitus and aural fullness.

1.2 Epidemiology

Onset is usually between 35 and 60 years. It begins in one ear, but the other ear becomes involved in a substantial minority over the years.

1.3 Ménière's disease or Ménière's syndrome

Ménière's disease is idiopathic hydrops. *Ménière's syndrome*, or secondary hydrops, has an identifiable cause: head injury, viral labyrinthitis, syphilis, Cogan syndrome,

autoimmune disease, or *delayed hydrops* years after a profound loss in that ear.

2 CHAPTER TWO Pathology

Too much endolymph.

2.1 Hydrops

The cochlear duct and saccule distend, bowing Reissner's membrane into the scala vestibuli. Small ruptures let potassium-rich endolymph mix with perilymph, poisoning hair cells and nerve fibres and producing an attack; as the membrane heals, function partly recovers.

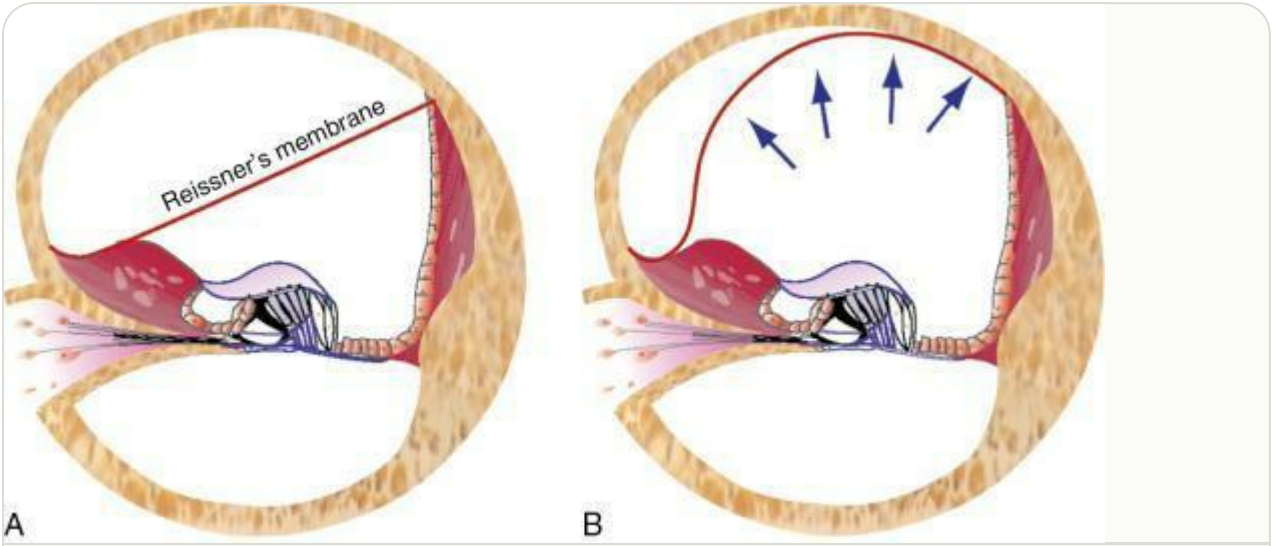


FIGURE 1 (A) The normal cochlear duct. (B) In hydrops the duct is distended with endolymph and Reissner's membrane bulges into the scala vestibuli.
Dhingra, fig. 15.1

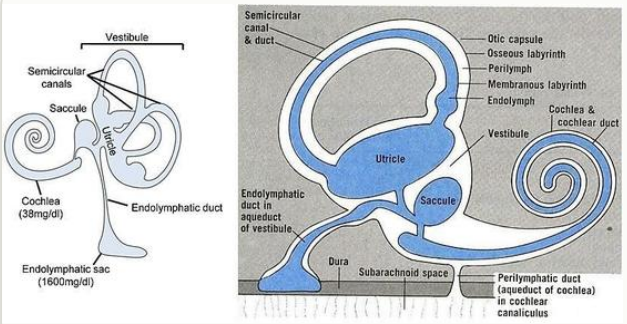


TABLE 2 Differences between the composition of endolymph, perilymph and CSF**

	Endolymph	Perilymph	CSF
Na^+ (mEq/L)	3	150	152
K^+ (mEq/L)	150	3–5	4
Cl^- (mEq/L)	130	125	
Protein (mg/dL)	126	200–400	20–50
Glucose (mg/dL)	10–40	85	70

**Values vary from the site of collection such as cochlea, saccule and endolymphatic sac in cases of endolymph and scala tympani and vestibuli in cases of perilymph.

FIGURE 2 The membranous labyrinth with the endolymphatic sac, and the composition of endolymph and perilymph.

Anatomy atlas · ENT Steps p. 74, ENT Steps p. 75

2.2 Why it happens

The cause is unknown; several theories may each be true in some patients.

- **Defective absorption** by the endolymphatic sac, or a small, poorly pneumatised sac and duct.
- **Vasomotor disturbance** with spasm of the internal auditory artery.
- **Allergy** and **autoimmunity.**
- **Salt and water retention.**
- **Hypothyroidism.**
- **Viral infection.**

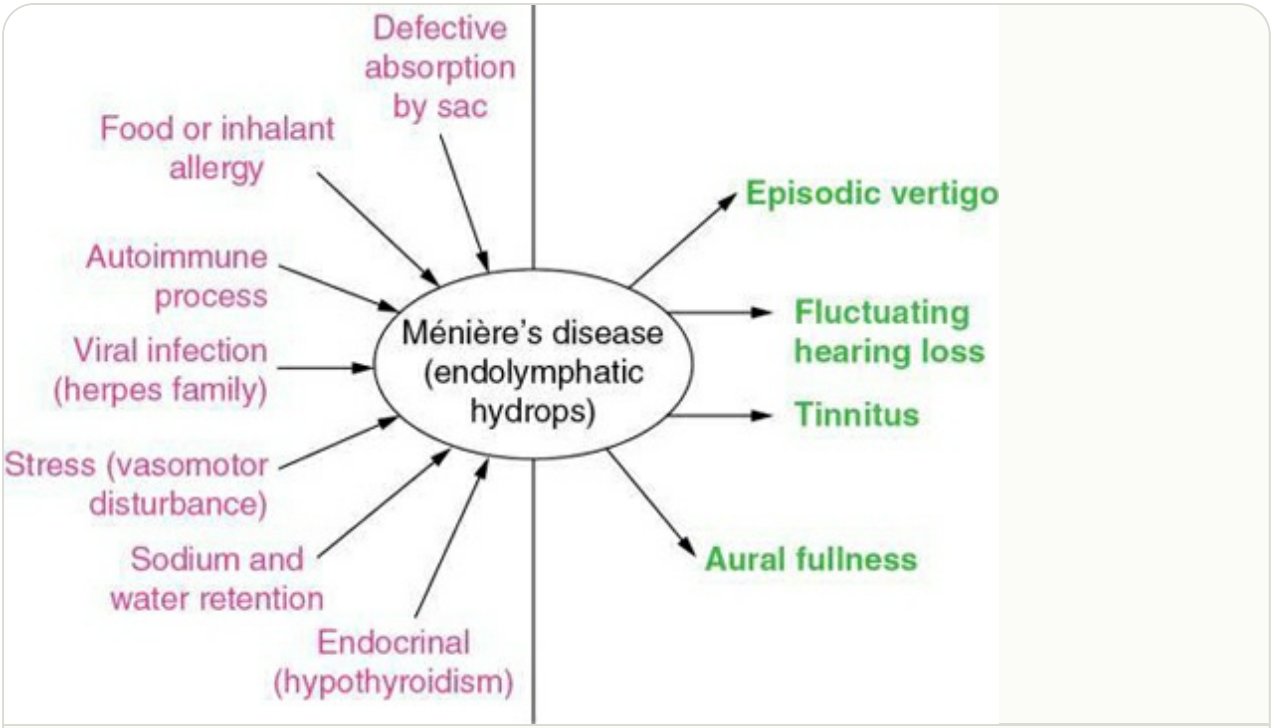


FIGURE 3 Aetiological factors and symptoms of endolymphatic hydrops.
Dhingra, fig. 15.2

3 CHAPTER THREE Clinical features

The attack and the interval.

3.1 Vertigo

Sudden, rotatory vertigo lasting from 20 minutes to several hours, often with nausea, vomiting, sweating and pallor. Attacks come in clusters with remissions of weeks or months. The patient lies still with the bad ear up or down; there is no loss of consciousness.

3.2 Hearing loss

Fluctuating and sensorineural, worse during an attack and recovering afterwards, at first in the low frequencies. Over the years the loss becomes permanent and flat. Distortion, *diplacusis* (hearing the same tone at a different pitch in each ear) and intolerance of loud sound are common.

3.3 Tinnitus

Low-pitched and roaring, louder before and during attacks.

3.4 Fullness

Pressure in the ear, which may change or worsen before an attack and act as a warning.

3.5 Signs

The drum is normal. During an attack there is horizontal-rotatory nystagmus.

WHERE THE BOOKS DIFFER · DIRECTION OF NYSTAGMUS IN AN ATTACK

Dhingra	Lee
Fast phase towards the healthy ear.	Towards the affected ear at the start (irritative), away from it later (paretic), and back towards it during recovery.

TULLIO PHENOMENON

Vertigo provoked by loud sound may occur when the distended saccule lies against the stapes footplate.

3.6 Variants

3.6.1 Cochlear hydrops

Hearing symptoms without vertigo.

3.6.2 Vestibular hydrops

Vertigo without hearing symptoms.

3.6.3 Lermoyez syndrome

Hearing gets worse before an attack and improves with the vertigo, the reverse of the usual pattern.

3.6.4 Tumarkin's otolithic crisis

> Sudden drop attacks without warning or loss of consciousness, from abrupt otolith stimulation. Dangerous falls make it an indication for definitive treatment.

4 CHAPTER FOUR Diagnosis

Criteria, audiogram, and exclusion.

4.1 Diagnostic criteria

AAO-HNS DIAGNOSTIC CATEGORIES

CATEGORY	REQUIREMENTS
Certain	Definite disease plus histological confirmation
Definite	Two or more spontaneous episodes of vertigo of 20 minutes or longer; documented hearing loss on at least one occasion; tinnitus or fullness; other causes excluded
Probable	One definite episode, documented hearing loss, tinnitus or fullness
Possible	Episodic vertigo without documented loss, or fluctuating loss with imbalance

BÁRÁNY SOCIETY CRITERIA (2015)

The criteria now used internationally, and by the Oxford Textbook, recognise only two categories. **Definite** Ménière's disease needs two or more spontaneous episodes of vertigo each lasting 20 minutes to 12 hours; audiometrically documented low- to medium-frequency sensorineural loss in the affected ear on at least one occasion before, during or after an attack; fluctuating aural symptoms (hearing, tinnitus or fullness) in that ear; and no better explanation. **Probable** disease needs two or more episodes of 20 minutes to 24 hours with fluctuating aural symptoms, without the audiometric requirement. The upper limit of 12 hours helps separate Ménière's disease from vestibular migraine, which is about ten times more common, overlaps with it in many patients, and is the commonest misdiagnosis.

4.2 Staging

By the average of thresholds at 0.5, 1, 2 and 3 kHz in the worst audiogram of the previous six months: stage 1 up to 25 dB, stage 2 26 to 40 dB, stage 3 41 to 70 dB,

stage 4 over 70 dB.

4.3 Audiometry

4.3.1 Pure-tone audiogram

Sensorineural loss affecting the low frequencies, giving a rising curve; later the curve flattens.

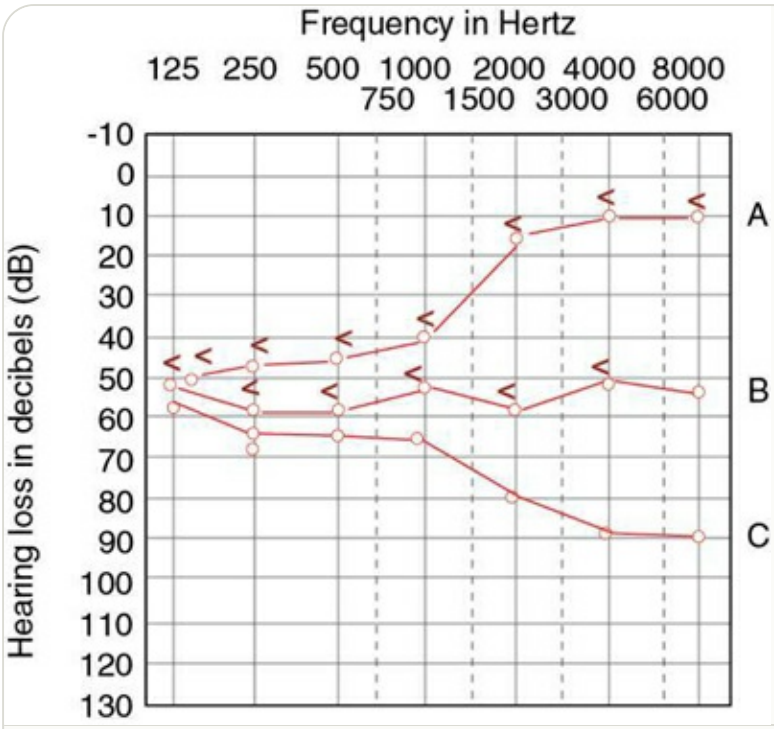


FIGURE 4 Audiogram in early Ménière's disease: sensorineural loss, worse in the low frequencies (a rising curve).

Dhingra, fig. 15.3

4.3.2 Speech and recruitment

Discrimination is moderately reduced, around 55 to 85% between attacks, and recruitment is present, marking a cochlear lesion.

4.4 Electrocochleography

The summating potential grows relative to the action potential in a hydropic ear.

WHERE THE BOOKS DIFFER · RAISED SP/AP RATIO

Dhingra	Lee
Above 30%.	Above 0.35.

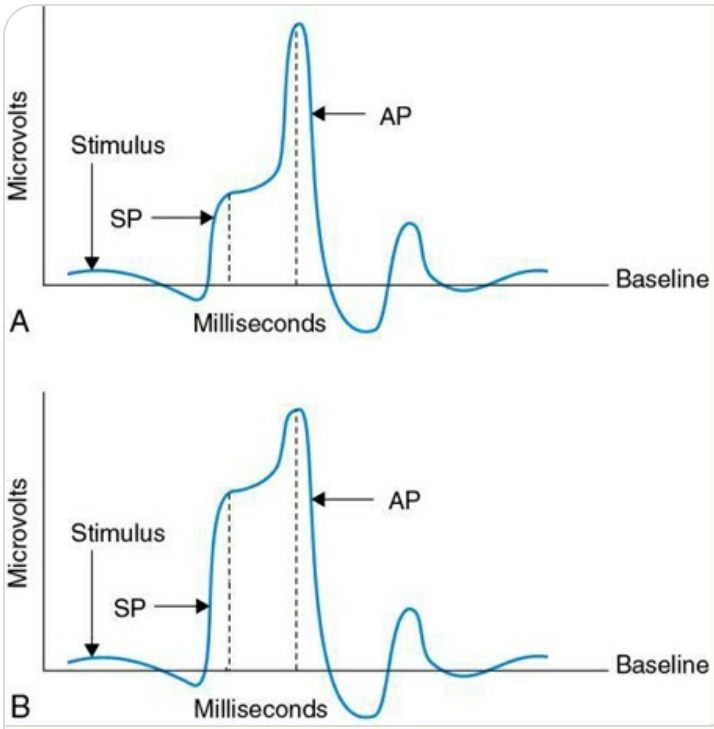


FIGURE 5 Electrocochleography: (A) normal ear, (B) Ménière's disease with an enlarged summating potential relative to the action potential.

Dhingra, fig. 15.4

4.5 Glycerol test

An osmotic diuretic, 1.5 mL/kg, reduces hydrops temporarily. An improvement of 10 dB at two adjacent frequencies, or 10% in discrimination, is positive.

4.6 Caloric test

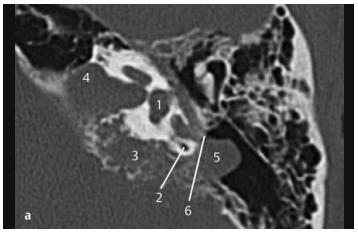
Reduced response on the affected side in about three-quarters.

4.7 MRI with contrast

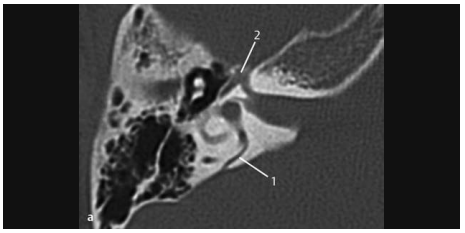
Essential to exclude a vestibular schwannoma or other retrocochlear lesion. Delayed contrast MRI can now show hydrops directly in specialist centres.

IMAGING IN THE APP 2 cases

An endolymphatic sac tumour and an enlarged vestibular aqueduct, two causes of secondary hydrops.



Endolymphatic Sac Tumour
Tumours · 8 images



Vestibular Aqueduct, Enlargements
Congenital · 11 images

4.8 Differential diagnosis

- Vestibular schwannoma.
- Vestibular migraine.
- Recurrent vestibulopathy.
- Superior canal dehiscence.
- Autoimmune inner ear disease and otosyphilis.
- Transient ischaemic attacks in the vertebrobasilar territory.

5 CHAPTER FIVE Management

Most are controlled medically.

5.1 The acute attack

AN ACUTE ATTACK

1. Reassure; let the patient lie still with the head supported.
2. Check the neurology and the head impulse test: new neurological signs mean stroke imaging, not sedatives.
3. Give an antiemetic or vestibular sedative: prochlorperazine, dimenhydrinate, or diazepam.

4 Intravenous fluids if vomiting.

5 Keep sedatives short: they slow compensation.

5.2 Long-term medical treatment

5.2.1 Lifestyle

Salt restriction to about 1.5 to 2 g a day, adequate water, limiting caffeine and alcohol, stopping smoking, and managing stress. Avoid working at heights or driving during active periods.

5.2.2 Medicines

Betahistine (with caution in asthma) and a diuretic. Treat allergy and hypothyroidism. About 80% are controlled.

WHERE THE BOOKS DIFFER · BETAHISTINE

Dhingra	Oxford
Betahistine and a diuretic as the standard long-term medical treatment.	A large randomised trial (BEMED, 2016) found no benefit of betahistine over placebo in reducing attacks, even at high dose. Many units still offer it as a low-risk first step, but patients with frequent attacks should not be held on it for long before moving to intratympanic steroids. Trials of intratympanic gentamicin and steroids show similar rates of hearing loss with either. The other ear's vestibular function must be checked before any ablation, because up to half of patients develop disease in the second ear over twenty years.

5.3 When medical treatment fails

The choice depends on whether hearing in the ear is still useful.

INTRACTABLE VERTIGO

HEARING USEFUL **Preserve the labyrinth** intratympanic steroid, endolymphatic sac surgery, pressure-pulse device

HEARING USEFUL **Destroy vestibular function** intratympanic gentamicin, vestibular neurectomy

HEARING LOST **Labyrinthectomy** removes all inner ear function

5.3.1 Intratympanic steroid

Repeated injections of dexamethasone or methylprednisolone; no risk to hearing.

5.3.2 Intratympanic gentamicin

Chemical ablation of the vestibular hair cells, given in small doses through a grommet or by injection. Vertigo is controlled in 60 to 80%, at the price of some hearing loss in 4 to 30%.

5.3.3 Pressure-pulse therapy

A device (Meniett) delivers low-pressure pulses to the middle ear through a grommet.

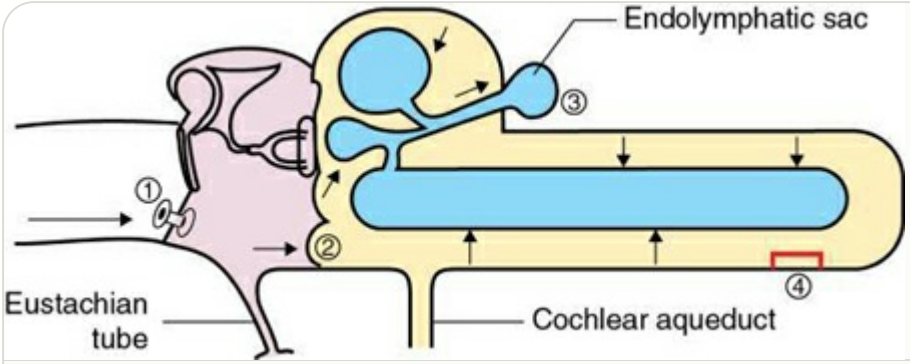


FIGURE 6 The mechanism of intermittent low-pressure pulse therapy.
Dhingra, fig. 15.6

5.3.4 Endolymphatic sac surgery

Decompression of the sac, or a shunt to the mastoid, preserves hearing. Its benefit over placebo is debated.

WHERE THE BOOKS DIFFER · SAC SURGERY

Dhingra	Lee
Listed among the operations that preserve hearing.	Controversial.

5.3.5 Vestibular neurectomy

Section of the vestibular nerve through the middle or posterior cranial fossa: high vertigo control with hearing preserved, at the price of a craniotomy.

5.3.6 Labyrinthectomy

For an ear with no useful hearing: removal of the membranous labyrinth through the canal or the mastoid. A cochlear implant can follow in the same or another sitting.

PEARLS

- Recruitment makes hearing aids hard to fit.
- Sacculotomy and the Cody tack are historical.
- Before destroying one ear, make sure the other ear is healthy.

OPERATIVE GUIDES IN THEATRE 2 operations

Surgery for Vertigo
Surgery for vertigo

Otitis Media, Myringotomy, Tympanostomy Tube, and Balloon Dilation
Tube for intratympanic or Meniett therapy

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The tetrad,** and the pathology of hydrops.
- **AAO-HNS criteria** and staging.
- **Audiogram:** low-frequency, fluctuating, rising curve.
- **Electrocochleography** and the glycerol test.
- **Variants:** Lermoyez and Tumarkin.
- **Ladder of treatment** by hearing status.

6.2 Common pitfalls

COMMON PITFALLS

- Labelling every dizzy patient as Ménière's without documented hearing loss.
- Forgetting MRI to exclude a schwannoma.
- Keeping patients on vestibular sedatives for months.
- Using gentamicin when the other ear may also be affected.

Written for this app from Oxford Textbook of Otolaryngology ch. 14, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 7, 15 and K. J. Lee ch. 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

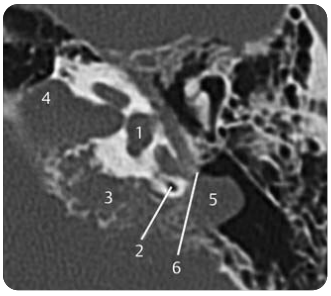
On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · TUMOURS

Endolymphatic Sac Tumour

- Signs of bone destruction in the region of the jugular bulb and jugular foramen. –
- Cranial nerve VII XI palsies, extension into structures of the neck and skull base.
- An endolymphatic sac tumor has a high risk of progressive morbidity and surgical complications due to excessive bleeding and cranial nerve damage. Preoperative embolization is recommended if technically possible.
- There is a well-known higher incidence of endolymphatic sac tumor in von Hippel–Lindau disease.
- Endolymphatic sac tumor arising from the middle third of the endolymphatic sac. It is a hypervascular tumor that may show intratumorous calcification. Bone destruction is characteristic, especially in patients with known von Hippel–Lindau disease.
- Remember to consider the possibility of bone destruction in the region of the jugular bulb and jugular foramen and the anticipated clinical symptoms with growth of the tumor when discussing the remaining surgical options with the patient, in particular the effects on cranial nerves VII XI.
- An endolymphatic sac tumor is associated with a high risk of progressive morbidity and surgical complications because of excessive bleeding and cranial nerve damage. Preoperative embolization is recommended if possible.



CT, axial. Posterior to the vestibule (1) and posterior semicircular canal (2), an opacification of the mastoid cells with infiltrative characteristics is seen. Thi…

INNER EAR & IAC · CONGENITAL

Vestibular Aqueduct, Enlargements

IMAGING FINDINGS

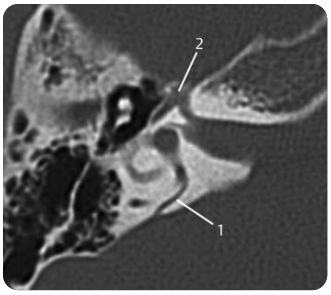
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

TOP DIFFERENTIALS

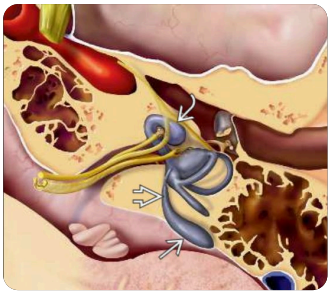
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

CHECKLIST

- Look at both ears: often bilateral
- Pendred syndrome association



CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.